

ECE-111 Advanced Digital Design Projects

Homework-2a:

Convert 1-bit ALU to N-bit ALU and change ALU module instantiation in ALU top module from explicit name port connection to dot-star port connection

- Port name of module alu in alu.sv can be changed when converting explicit name port connection to dot-star port connection for alu instantiation in module alu_top.sv
- Synthesize alu_top and alu module
- Using provided alu testbench, run simulation for dot-star based alu instantiation in alu_top
- Review synthesis results (FPGA resource usage and RTL netlist/schematic)
- Review input and output signals in simulation waveform

Homework Submission :

Submit report (PDF file) which should include :

- alu_top, alu SystemVerilog design and testbench code
- Explain changes done in SystemVerilog code to change it from explicit name based port connections to dot-start based port connections for alu instantiation
- Explain changes done in SystemVerilog code to convert 1-bit ALU to 4-bit ALU
- Review post-Synthesis FPGA resource usage and schematic generated from RTL netlist viewer
- Simulation snapshot, transcript snapshot and explain simulation result

Note :

Lab2.zip folder has **alu_top.sv**, **alu.sv** and **alu_top_testbench.sv** which should be used as a starting point for Homework-2a.

Homework-2b:

Synthesis and Simulate N-bit counter SystemVerilog Code

- Review synthesis results (resource usage and RTL netlist/schematic)
- Run simulation using counter testbench code
- Review counter input output signals in simulation waveform

Homework Submission :

Submit report (PDF file) which should include :

- Counter SystemVerilog and testbench code
- Post synthesis FPGA resource usage and schematic generated from RTL netlist viewer
- Simulation snapshot, transcript snapshot and explain simulation result

Note :

Lab2.zip folder has **counter_4bit.sv** and **counter_4bit_testbench.sv** which should be used for Homework-2b.